# Question No:2

2) Semantic Network Using NetworkX Aim: To represent concepts and labeled relationships as a directed semantic network. Software Required: Python 3.x and NetworkX (install with: pip install networkx)

## Algorithm: Semantic Network Using NetworkX

1. Start.
2. Import the networkx library.
3. Create a directed graph.
4. Add concepts as nodes through edges.
5. Add relationships between concepts, such as is-a and has-instance.
6. Display each relationship between the concepts.
7. Draw the semantic network graph.
8. Display the graph with labeled nodes and edges.
9. Stop.

## Program:

```
!pip install networkx matplotlib

import networkx as nx

import matplotlib.pyplot as plt

# Create a directed graph

G = nx.DiGraph()

# Add concepts and relationships

G.add_edge("Dog", "Mammal", relation="is-a")

G.add_edge("Mammal", "Animal", relation="is-a")

G.add_edge("Dog", "Tommy", relation="has-instance")

# Display relationships

print("Semantic Network Relationships:")

for source, target, data in G.edges(data=True):
```

```
    print(source, "--", data["relation"], "-->", target)
# Draw the semantic network
pos = nx.spring_layout(G, seed=42)
nx.draw(G, pos, with_labels=True, node_color="lightblue",
        node_size=2500, font_size=10, arrows=True)
edge_labels = nx.get_edge_attributes(G, "relation")
nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels)
plt.title("Semantic Network")
plt.show()
```

OUTPUT:

```
Semantic Network Relationships:
Dog -- is-a --> Mammal
Dog -- has-instance --> Tommy
Mammal -- is-a --> Animal
```

